Database Overview

The database is designed to manage a faculty portfolio system. It consists of 10 interconnected tables tracking users, profiles, departments, courses, and various faculty accomplishments (achievements, education, experience, projects, and publications).

Table Relationships

- **users** (1) to (1) **profiles**: A user account is linked to a faculty profile.
- **departments** (1) to (Many) **courses**: A department can offer multiple courses.
- **departments** (1) to (Many) **profiles**: A department can have multiple faculty members.
- **profiles** (1) to (Many) **achievements**: A faculty member can have multiple achievements.
- **profiles** (1) to (Many) **education**: A faculty member can have multiple educational backgrounds.
- **profiles** (1) to (Many) **experience**: A faculty member can have multiple past experiences.
- **profiles** (1) to (Many) **faculty_courses**: A faculty member can teach multiple courses.
- **courses** (1) to (Many) **faculty_courses**: A course can be taught by multiple faculty members across semesters.
- **profiles** (1) to (Many) **projects**: A faculty member can lead or participate in multiple projects.
- **profiles** (1) to (Many) **publications**: A faculty member can author multiple publications.

Table Data Dictionaries

1. users

Stores system authentication and basic user account details.

- **id**: uuid (Primary Key, Default: uuid_generate_v4())
- **email**: character varying(255) (Unique, Not Null)
- **password_hash**: character varying(255) (Not Null)
- **role**: character varying(20)

- **created_at**: timestamp without time zone (Default: CURRENT_TIMESTAMP)

2. profiles

Stores detailed information about faculty members.

- **id**: uuid (Primary Key, Default: uuid_generate_v4())
- **user_id**: uuid (Foreign Key referencing users.id)
- **department_id**: uuid (Foreign Key referencing departments.id)
- **full_name**: character varying(255)
- **designation**: character varying(100)
- **bio**: text
- **phone**: character varying(20)
- **office_location**: character varying(100)
- **avatar_url**: character varying(255)
- **google_scholar_link**: character varying(255)
- **orcid**: character varying(100)
- **is_verified**: boolean (Default: false)

3. departments

Stores academic department or college classifications.

- **id**: uuid (Primary Key, Default: uuid_generate_v4())
- **name**: character varying(100) (Not Null)
- **college**: character varying(100)
- **created_at**: timestamp without time zone (Default: CURRENT_TIMESTAMP)

4. courses

Stores the catalog of courses available in the institution.

- **id**: uuid (Primary Key, Default: uuid_generate_v4())
- **course_code**: character varying(20)
- **course_name**: character varying(255)
- **department_id**: uuid (Foreign Key referencing departments.id)
- **credits**: integer

5. faculty_courses

Junction table tracking which courses are taught by which faculty members per semester.

- **id**: uuid (Primary Key, Default: uuid_generate_v4())
- **profile_id**: uuid (Foreign Key referencing profiles.id)
- **course_id**: uuid (Foreign Key referencing courses.id)
- **semester**: character varying(20)
- **year**: integer

6. achievements

Stores awards, honors, and other recognitions received by faculty members.

- **id**: uuid (Primary Key, Default: uuid_generate_v4())
- **profile_id**: uuid (Foreign Key referencing profiles.id)
- **title**: character varying(255)
- **description**: text
- **awarded_date**: date
- **issuing_organization**: character varying(255)

7. education

Stores the academic background and degrees earned by faculty members.

- **id**: uuid (Primary Key, Default: uuid_generate_v4())
- **profile_id**: uuid (Foreign Key referencing profiles.id)
- **degree**: character varying(100)
- **field_of_study**: character varying(100)
- **university**: character varying(255)
- **start_year**: integer
- **end_year**: integer

8. experience

Stores the professional and academic work history of faculty members.

- **id**: uuid (Primary Key, Default: uuid_generate_v4())
- **profile_id**: uuid (Foreign Key referencing profiles.id)

- **institution**: character varying(255)
- **position**: character varying(100)
- **start_date**: date
- **end_date**: date

9. projects

Stores information about research projects and grants.

- **id**: uuid (Primary Key, Default: uuid_generate_v4())
- **profile_id**: uuid (Foreign Key referencing profiles.id)
- **title**: character varying(255)
- **description**: text
- **funding_agency**: character varying(255)
- **grant_amount**: numeric
- **start_date**: date
- **end_date**: date
- **status**: character varying(20)

10. publications

Stores research papers, articles, and other academic works.

- **id**: uuid (Primary Key, Default: uuid_generate_v4())
- **profile_id**: uuid (Foreign Key referencing profiles.id)
- **title**: character varying(255)
- **authors**: text
- **journal_conference**: character varying(255)
- **publication_type**: character varying(50)
- **publish_date**: date
- **doi**: character varying(100)
- **abstract**: text
- **document_url**: character varying(255)